

PREDIVISEUR MB506 (division fixe 1/64)

MISE EN FORME 3V -> D5

ARDUINO PRO MINI (alimente en 3V)

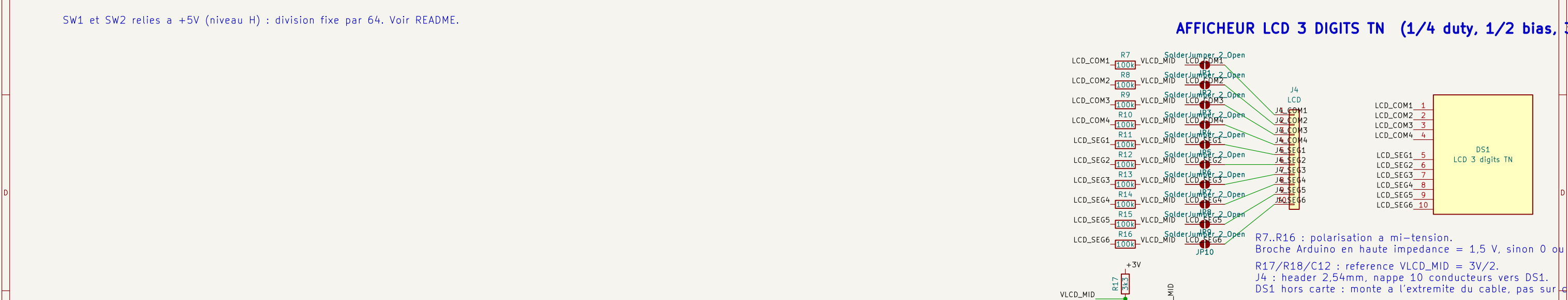
The schematic diagram illustrates the circuit for the 'MISE EN FORME 3V -> D5' project. It features a PREDIVISEUR MB506 (division fixe 1/64) and an ARDUINO PRO MINI (alimente en 3V). The circuit includes various components such as resistors (R3, R4, R6, R19, R20), capacitors (C7, C8, C9, C10, C11), and a 2SC5551AE-TD-E Q1 transistor. The PREDIVISEUR MB506 is connected to a +5V supply and its output is connected to the D5_SIG pin of the ARDUINO PRO MINI. The ARDUINO PRO MINI is powered by a 3V supply and its output is connected to the D5_SIG pin of the PREDIVISEUR MB506. The circuit also includes a 100nF capacitor (C11) connected to the 3V supply and a 100nF capacitor (C9) connected to the +5V supply.

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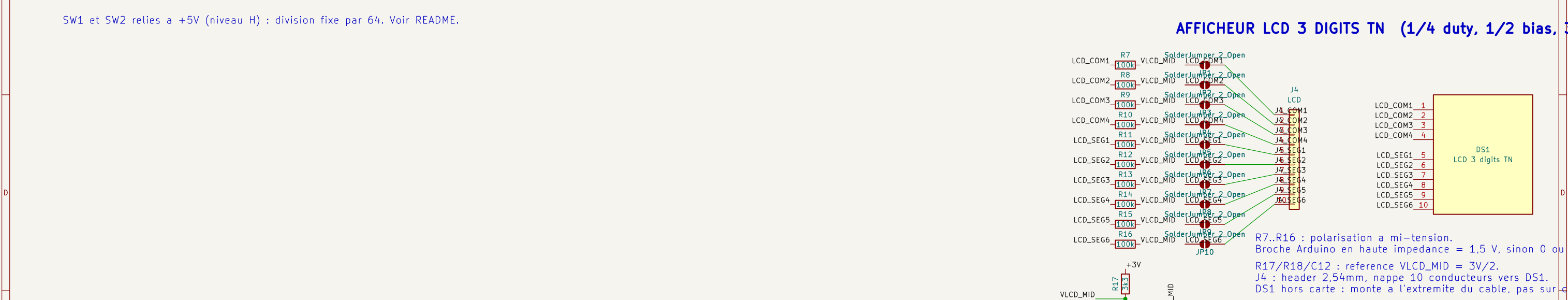
The diagram illustrates a 3V to 5V level shifter circuit. It consists of a fixed divider (MB506) and a 2SC5551AE-TD-E transistor (Q1). The divider is connected to the 5V supply and ground, with a 100nF capacitor (C8) and a 1nF capacitor (C7) for filtering. The transistor's base is connected to the divider output, and its emitter is connected to ground. The collector is connected to the 3V supply and ground, with a 100nF capacitor (C9) and a 1nF capacitor (C10) for filtering. The output signal (D5_SIG) is taken from the collector. The circuit is powered by a 5V supply and a 3V supply.

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The Arduino Pro Mini is shown with its pin connections. The 3V pin is connected to the 3V supply. The GND pin is connected to ground. The D5_SIG pin is connected to the collector of the transistor. The LCD segment pins (LCD_SEG1 to LCD_SEG6) are connected to the corresponding pins on the Arduino Pro Mini.



SW1 et SW2 relies a +5V (niveau H) : division fixe par 64. Voir README.

AFFICHEUR LCD 3 DIGITS TN (1/4 duty, 1/2 bias, 10 segments)

The schematic shows the following components and connections:

- Resistors:** R7-R16 are 100k resistors connected to VLCD_MID. R17 is a 3k3 resistor connected to +3V.
- Solder Jumpers:** Labeled "SolderJumper 2 Open".
- Header J4:** A 10-pin header with pins J4_COM1 through J4_SEG6.
- LCD Module DS1:** A yellow box representing the LCD module with pins 1-10 corresponding to the header pins.

Pinout details from the diagram:

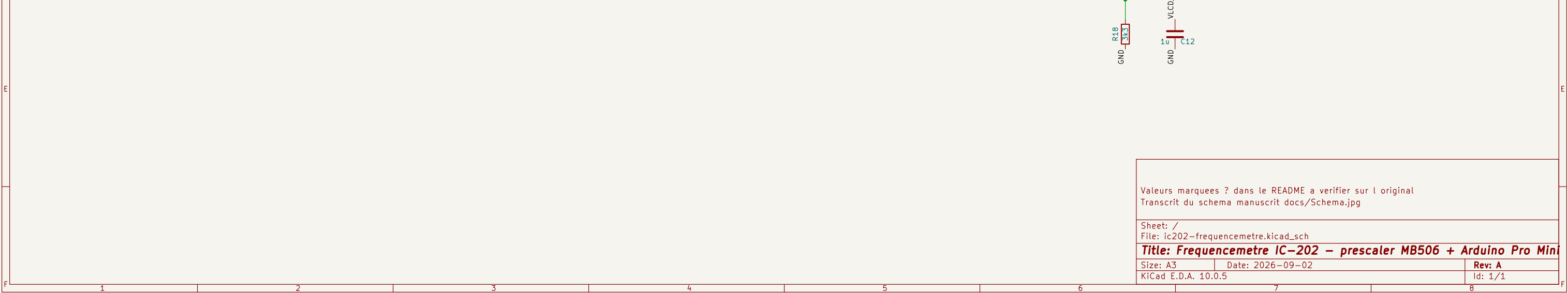
Label	Value / Component	Connection
R7	100k	VLCD_MID to LCD_COM1
R8	100k	VLCD_MID to LCD_COM2
R9	100k	VLCD_MID to LCD_COM3
R10	100k	VLCD_MID to LCD_COM4
R11	100k	VLCD_MID to LCD_SEG1
R12	100k	VLCD_MID to LCD_SEG2
R13	100k	VLCD_MID to LCD_SEG3
R14	100k	VLCD_MID to LCD_SEG4
R15	100k	VLCD_MID to LCD_SEG5
R16	100k	VLCD_MID to LCD_SEG6
R17	3k3	+3V to VLCD_MID

Connections to the LCD module (DS1):

- J4_COM1 to LCD_COM1
- J4_COM2 to LCD_COM2
- J4_COM3 to LCD_COM3
- J4_COM4 to LCD_COM4
- J4_SEG1 to LCD_SEG1
- J4_SEG2 to LCD_SEG2
- J4_SEG3 to LCD_SEG3
- J4_SEG4 to LCD_SEG4
- J4_SEG5 to LCD_SEG5
- J4_SEG6 to LCD_SEG6

Additional notes from the diagram:

- R7..R16 : polarisation a mi-tension. Broche Arduino en haute impédance = 1,5 V, sinon 0 ou 5V.
- R17/R18/C12 : reference VLCD_MID = 3V/2.
- J4 : header 2,54mm, nappe 10 conducteurs vers DS1.
- DS1 hors carte : monte à l'extrémité du câble, pas sur la carte.



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